1 ***Planning (15 marks)***

Defining the problem (3 marks)

P1 *r* is the independent variable or vary *r* (accept diameter but not mass or size). [1]

P2 *v* is the dependent variable or determine *v* (accept speed) [1]

P3 A controlled variable – accept temperature, [1]
distance when time is measured, or time when distance measured.
Do not accept volume/height of oil.

Methods of data collection (5 marks)

M1 Diagram of a workable arrangement including a deep container of oil, ball and some measurement indicated for either time or distance. [1]

M2 Measure diameter by using a micrometer (screw gauge)/vernier callipers (and halving to obtain radius). Accept from diagram. Accept travelling microscope. [1]

M3 Measure the time for the ball to fall a set distance in oil (or distance for a set time). [1]

M4 Measure the (constant) distance fallen (constant time) and show how *v* is calculated. [1]

M5 Evidence that ball has reached terminal velocity (e.g. starting mark well below surface of oil) [1]
Reject equations of uniform acceleration ideas.

Method of analysis (2 marks)

A1 Plot a graph of *v* against r^2 or logarithmic equivalent. [1]

A2 Relationship is correct if graph is a straight line through the origin. [1]
An explicit statement is required.
If lg v against lg r is plotted gradient should equal 2.

Safety considerations (1 mark)

S1 Relevant safety precaution related to the oil, [1]
e.g. mop up spillages of oil/wear gloves with reason/keep away from flames.
Do not accept vague answers e.g. goggles/spills/washing hands but allow credit for detailed reasoning e.g. drop ball near surface to avoid splashing.

Additional detail (4 marks)

D1/2/3/4 Relevant points might include: [4]
Allow oil to stand so that air bubbles escape/ball may trap air bubbles.
Wash and dry steel balls/handle steel balls with tweezers/gloves.
Distance marks should be as far apart as possible or use long tube.
Large distance to reduce percentage uncertainty.
Wide tube to reduce edge effects/method to keep long tube vertical.
Discussion of parallax for stop watch methods.
Method of ensuring that terminal velocity has been reached.
Retrieve steel balls using a magnet.
Use clear oil.
Repeat diameter measurements and average.
An additional variable kept constant.

[Total: 15]

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